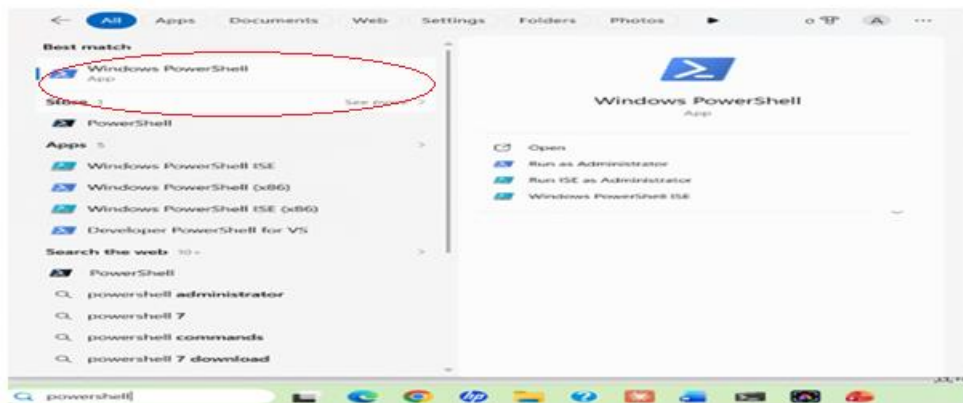


Skill1:

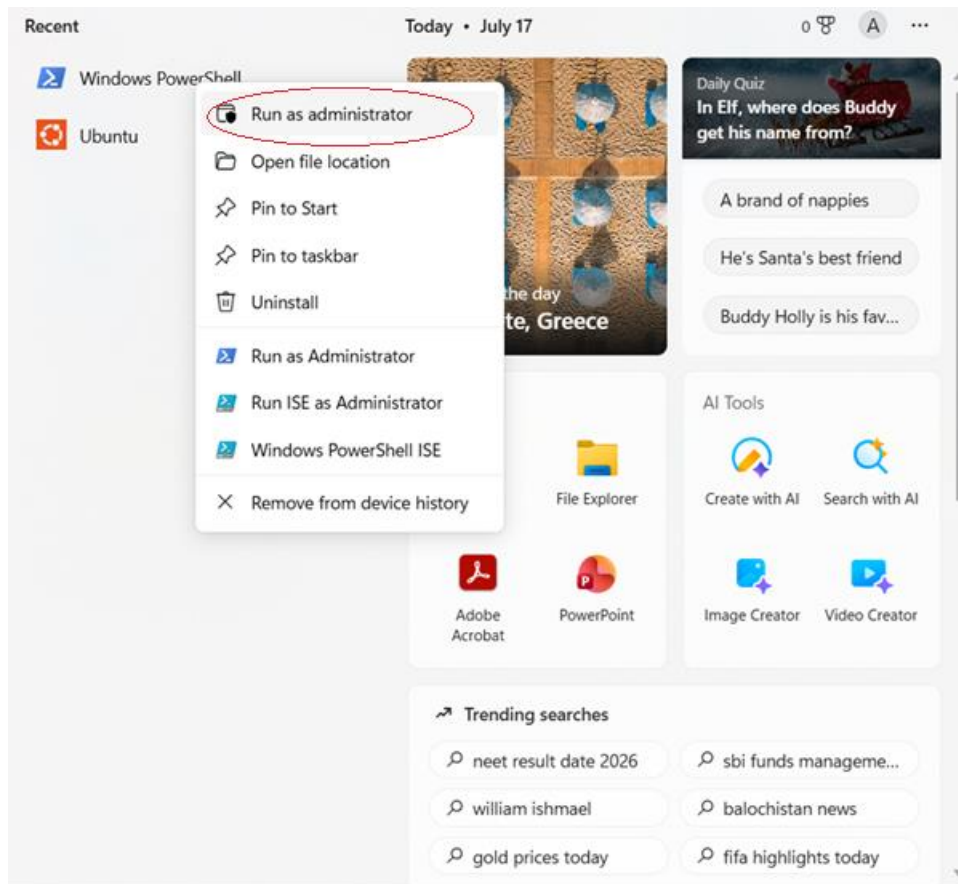
1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\windows\system32> wsl --install
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: anusha

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

Step 6: Update Ubuntu

```
manisha@LAPTOP-5SMR05RJ:~$ sudo apt update
[sudo: password for manisha:]
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [293 kB]
Hit:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [357 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [76.5 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [31.1 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [91.1 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [138 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [53.1 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [153 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [43.9 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [25.5 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [24.5 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [47.6 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [45.5 kB]
Fetched 2140 kB in 7s (295 kB/s)
74 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Step 7: Install GCC (C Compiler)

```
manisha@LAPTOP-5SMR05RJ:~$ sudo apt install build-essential
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  build-essential
0 upgraded, 1 newly installed, 0 to remove and 74 not upgraded.
Need to get 4,758 B of archives.
After this operation, 20.5 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 build-essential amd64 12.10ubuntu1 [4,758 B]
Fetched 4,758 B in 1s (6,201 B/s)
Selecting previously unselected package build-essential.
(Reading database ... 215000 files and directories currently installed.)
Preparing to unpack .../build-essential_12.10ubuntu1_amd64.deb ...
Unpacking build-essential (12.10ubuntu1) ..
Setting up build-essential (12.10ubuntu1) ...
Processing triggers for man-db (2.12.0-4build2) ...
manisha@LAPTOP-5SMR05RJ:~$
```

Step 8: Verify the Installation

```
manisha@LAPTOP-5SMR05RJ:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
manisha@LAPTOP-5SMR05RJ:~$
```

Step 9: Create Your C Program

Example: nano process.c

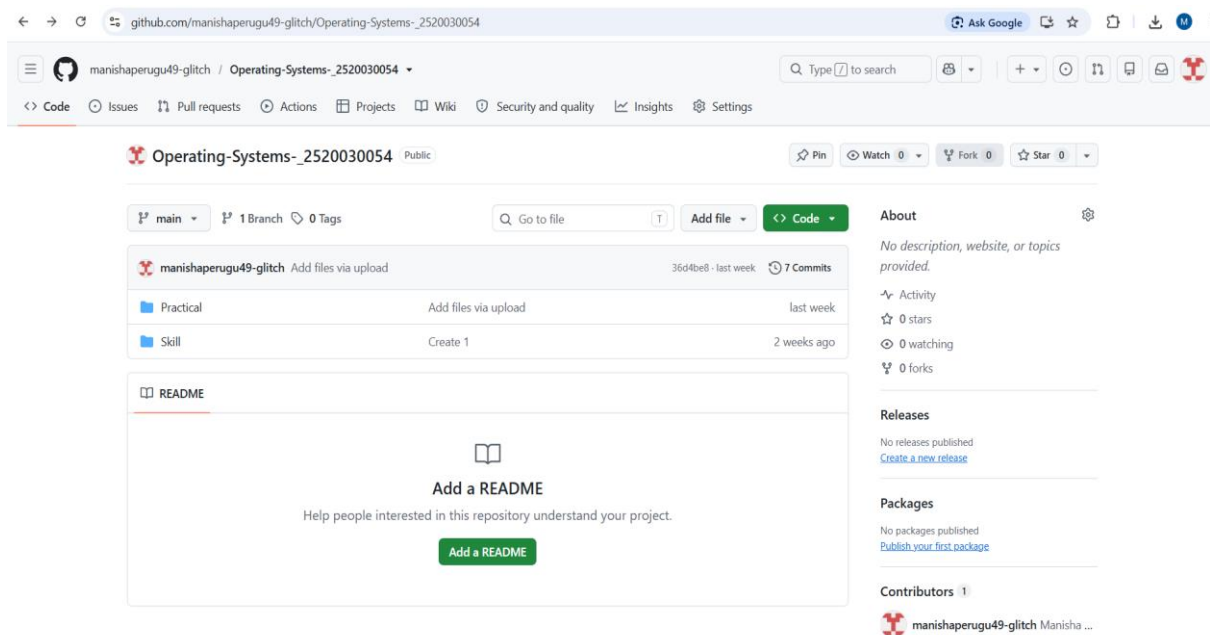
Step 10: Compile the Program

```
manisha@LAPTOP-5SMR05RJ:~$ gcc process.c -o process
manisha@LAPTOP-5SMR05RJ:~$ ./process
Enter a Linux command: pwd

Parent Process
Parent PID : 1876
Child PID  : 1877

Child Process
Child PID  : 1877
Parent PID : 1876
/home/manisha
Child process completed.
manisha@LAPTOP-5SMR05RJ:~$
```

GITHUB:



To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp("ls", "ls", "-l", NULL);

        perror("exec failed");
        exit(1);
    }
    else {
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());

        wait(NULL);
        printf("Child process completed.\n");
    }

    return 0;
}
```

```

manisha@LAPTOP-5SMR05RJ:~/OSlab$ nano fork_exec_parent_child.c
manisha@LAPTOP-5SMR05RJ:~/OSlab$ gcc fork_exec_parent_child.c -o fork_exec_parent_child
manisha@LAPTOP-5SMR05RJ:~/OSlab$ ./fork_exec_parent_child
Parent Process
Parent PID: 1162
Child Process
Child PID: 1163
Parent PID: 1162
total 400
-rwxr-xr-x 1 manisha manisha 15992 Jul 30 04:10 ID
-rw-r--r-- 1 manisha manisha 99 Jul 30 04:09 ID.c
drwxr-xr-x 2 manisha manisha 4096 Jul 29 06:45 Test1
-rwxr-xr-x 1 manisha manisha 16312 Aug 5 08:35 command
-rw-r--r-- 1 manisha manisha 557 Aug 5 08:31 command.c
-rw-r--r-- 1 manisha manisha 557 Aug 5 08:30 command.c.save
-rwxr-xr-x 1 manisha manisha 16176 Aug 6 05:46 copyfile
-rw-r--r-- 1 manisha manisha 533 Aug 6 04:12 copyfile.c
-rwxr-xr-x 1 manisha manisha 16000 Aug 12 06:57 exec_demo
-rw-r--r-- 1 manisha manisha 179 Aug 12 06:57 exec_demo.c
-rw-r--r-- 1 manisha manisha 34 Aug 6 04:15 file1.txt
-rwxr-xr-x 1 manisha manisha 15992 Jul 30 04:16 fork
-rw-r--r-- 1 manisha manisha 96 Jul 30 04:15 fork.c
-rwxr-xr-x 1 manisha manisha 16312 Aug 13 05:36 fork_exec_parent_child
-rw-r--r-- 1 manisha manisha 622 Aug 13 05:35 fork_exec_parent_child.c
-rwxr-xr-x 1 manisha manisha 16096 Aug 13 04:55 fork_parent_child_pid
-rw-r--r-- 1 manisha manisha 296 Aug 13 04:55 fork_parent_child_pid.c
-rwxr-xr-x 1 manisha manisha 16224 Aug 13 05:06 fork_parent_child_wait
-rw-r--r-- 1 manisha manisha 399 Aug 13 05:03 fork_parent_child_wait.c
-rwxr-xr-x 1 manisha manisha 16088 Aug 12 07:00 fork_pid_loop
-rw-r--r-- 1 manisha manisha 198 Aug 12 06:59 fork_pid_loop.c
-rwxr-xr-x 1 manisha manisha 16080 Aug 13 04:51 forkwait
-rw-r--r-- 1 manisha manisha 335 Aug 12 06:50 forkwait.c
-rwxr-xr-x 1 manisha manisha 15952 Jul 30 03:45 hello
-rw-r--r-- 1 manisha manisha 85 Jul 30 03:56 hello.c
-rw-r--r-- 1 manisha manisha 6 Aug 6 04:16 input.txt
-rwxr-xr-x 1 manisha manisha 16128 Aug 13 05:08 multi_child_wait
-rw-r--r-- 1 manisha manisha 367 Aug 13 05:08 multi_child_wait.c
-rwxr-xr-x 1 manisha manisha 16056 Aug 12 06:54 multi_fork_children
-rw-r--r-- 1 manisha manisha 236 Aug 12 06:54 multi_fork_children.c
-rwxr-xr-x 1 manisha manisha 16176 Aug 12 07:06 orphanprocess
-rw-r--r-- 1 manisha manisha 314 Aug 12 07:05 orphanprocess.c
-rw-r--r-- 1 manisha manisha 6 Aug 6 05:46 output.txt
-rwxr-xr-x 1 manisha manisha 16040 Aug 1 05:57 parent
-rw-r--r-- 1 manisha manisha 222 Aug 1 05:56 parent.c
-rwxr-xr-x 1 manisha manisha 16000 Aug 12 06:42 processid
-rw-r--r-- 1 manisha manisha 112 Aug 12 06:41 processid.c
-rwxr-xr-x 1 manisha manisha 16056 Jul 30 04:03 username
-rw-r--r-- 1 manisha manisha 134 Jul 30 04:04 username.c
-rwxr-xr-x 1 manisha manisha 16032 Aug 1 06:01 wait
-rw-r--r-- 1 manisha manisha 207 Aug 1 06:01 wait.c
-rwxr-xr-x 1 manisha manisha 16040 Aug 12 07:03 zombieprocess
-rw-r--r-- 1 manisha manisha 254 Aug 12 07:03 zombieprocess.c
Child process completed.
manisha@LAPTOP-5SMR05RJ:~/OSlab$ |

```